FPT UNIVERSITY

Multi-Modal Fusion Techniques for Abnormal Events Detection in Videos

Ta Tran Thanh Tung
Vu Minh Hoang
Tran Minh Hoan
Dinh Cong Binh
Vuong Tu Binh

Bachelor of Artificial Intelligence

Supervisor: Dr. Bui Van Hieu

Major: Artificial Intelligence

University: FPT University

Hoa Lac campus, 04/2025

Contents

Acknowledgements . . . 3
Abstract . . . 4
List of Figures . . . 5
List of Tables . . . 6
1 Introduction . . . 7
2 Related Works . . . 7
3 Methods . . . 8
3.1 Audio Feature Extraction . . . 9
3.2 RGB and Flow Feature Extraction . . . 10
3.3 Pose Feature Extraction . . . 11
3.4 Unified Multi-Modal Anomaly Detection Framework . . . 13
3.5 Multimodal Alignment . . . 13
3.6 Gated Multimodal Fusion and Anomaly Detection . . . 15
3.7 Loss Formulation and Contrastive Loss . . . 17
4 Results And Discussion . . . 18
4.1 Dataset and Evaluation Metric . . . 18
4.2 Implementation Details . . . 19
4.3 Experimental Results on XD-Violence . . . 19
4.4 Qualitative Analysis . . . 21
5 Conclusion . . . 24
References . . . 24

Acknowledgements

We would like to express our deepest gratitude to Dr. Bui Van Hieu for his unwavering support, invaluable guidance, and continuous encouragement throughout the completion of this thesis. His profound expertise and insightful mentorship have not only shaped our academic journey but have also instilled in us the confidence and curiosity to pursue scholarly research with diligence and passion. His dedication to education and research has been a constant source of inspiration, and for that, we are truly grateful.

We also extend our heartfelt appreciation to FPT University for providing an outstanding academic environment, equipped with modern facilities and abundant resources that have greatly facilitated our learning and research. The knowledge and experiences gained during our time here have been instrumental in our intellectual and professional growth.

Beyond academic support, we are deeply indebted to our families, friends, and loved ones, whose unwavering encouragement and belief in us have been a pillar of strength. Their patience, sacrifices, and unconditional support have carried us through sleepless nights, moments of doubt, and the many challenges we encountered along the way. Without them, this journey would not have been the same.

As we reflect on this remarkable journey, we cherish the invaluable experiences, friendships, and lessons that have shaped us. Though our paths may diverge in the future, the bonds we have built and the knowledge we have acquired will remain with us as guiding lights.

Once again, we extend our sincerest gratitude to everyone who has played a part in our academic and personal development. This milestone is not solely our achievement but a testament to the collective support and kindness we have received.

Abstract

Detecting abnormal events in videos is crucial for enhancing public safety, yet existing methods often struggle to effectively integrate multimodal information such as appearance, motion, audio, and pose due to feature misalignment, redundancy, and inconsistent fusion. Traditional strategies like early, late, or attention-based fusion fail to ensure semantic coherence across modalities, limiting detection performance and generalization. In this study, we experiment a unified framework that incorporates four complementary data streams comprise RGB, optical flow, audio, and pose that come along with a Modality-wise Feature Matching Subspace (MFMS) to align features into a shared latent space, and a Gated Multimodal Fusion (GMF) mechanism to adaptively weigh modality contributions. A contrastive loss is used to further enhance semantic consistency. Our method is evaluated on the XD-Violence dataset and achieves an Average Precision (AP) of **84.98%**, demonstrating the effectiveness of our approach in capturing complex abnormal behaviors through structured alignment and fusion.

Keywords: Multimodal anomaly detection, feature fusion, cross-modal alignment, weak supervision, video surveillance.

List of Figures

Figure 1: Overview of the Multimodal Anomaly Detection Framework 9
Figure 2: Overview of the audio feature extraction and classification pipeline [19]. 10
Figure 3: Flow feature processing pipeline [20] 11
Figure 4: Pose feature processing pipeline. 13
Figure 5: MFMS for modality alignment. The process includes linear projection of each modality's features into a shared latent space followed by intra-modality refinement using self-attention and inter-modality interaction using cross-attention. 14
Figure 6: Illustration of the GMF process. Each modality feature f_c is first processed through a gating mechanism to compute the gating factor g_m, which determines its contextual importance. The original feature is then reweighted by the gating factor, and all reweighted modalities are aggregated to form the fused representation. 16
Figure 7: Residual Fusion Block incorporating skip connection and LayerNorm to stabilize and enrich the fused representation. 16
Figure 8: Structured combination of loss functions for potent optimization. 17
Figure 9: Contrastive loss mechanism for enhancing feature discriminability. 18
Figure 10: Training Loss and Accuracy Trends of Our Model. 22
Figure 11: Area Under the Curve (AUC) of AP over Iterations 22
Figure 12: Anomaly score timeline on an out-of-distribution UFC fight video. Pink areas mark manually labeled violent events. 23
Figure 13: Anomaly score timeline on a non-violent romantic scene video. 24

List of Tables

Table 1: Hyperparameter Setup . . . 19
Table 2: Performance Comparison on the XD-Violence Dataset . . . 20
Table 3: Training Metrics at Iteration 850 . . . 20
Table 4: Impact of Adding Pose Modality . . . 20
Table 5: Impact of Gated Multimodal Fusion . . . 21
Table 6: Impact of Contrastive Loss Integration . . . 21
Table 7: Impact of Modality-wise Feature Matching Subspace (MFMS) . . . 21

1 Introduction

Abnormal events refer to rare, unexpected and contextually deviant activities that differentiate themselves from normal behavior patterns in a given environment. These events are often indicative of danger, malfunction, or disruption, making their timely detection critical in real-world applications. With the explosive growth of video data in recent years, automatic abnormal event detection (AED) has become increasingly crucial across various domains. In security surveillance, the timely detection of unusual behaviors can help prevent crimes and ensure public safety [30]. In healthcare monitoring, identifying abnormal patient activities such as sudden falls or irregular movements can enable rapid intervention and improve patient outcomes. In industrial settings, detecting anomalies such as equipment malfunctions, fire outbreaks, or hazardous conditions can minimize damages and prevent catastrophic failures [17]. Furthermore, in traffic monitoring systems, the ability to recognize abnormal events, such as accidents or road blocks, contributes to efficient traffic management and accident prevention. Given the wide range of applications and the high stakes involved, the development of robust and efficient methods for AED in videos is of paramount importance.

Anomaly detection in videos has evolved significantly over the years, transitioning through several key stages. The early methods primarily relied on hand-made features, such as optical flow, trajectory modeling, and motion histograms [27, 6], coupled with classical machine learning algorithms. Although effective in controlled environments, these approaches struggled with the complexity and variability of real-world video data. The advent of deep learning marked a major breakthrough. Convolutional neural networks (CNN) captured rich spatial patterns, while repeat neural networks (RNNs) modeled temporal dependencies, leading to substantial performance improvements [14, 23]. However, limitations in modeling long-range interactions remained, motivating the adoption of transformer-based architectures [9]. Leveraging self-attention mechanisms, transformers enabled global context modeling and significantly improved anomaly detection in complex video scenes [33]. In parallel, the scarcity of labeled anomaly data spurred the rise of self-supervised learning strategies [1], where models are pre-trained on auxiliary tasks such as temporal ordering or contrastive learning. These approaches further enhanced feature generalization without heavy reliance on manual annotations, forming the foundation for more robust and scalable anomaly detection systems.

Despite remarkable progress, video anomaly detection still faces considerable challenges when deployed in real-world, multimodal environments. Hybrid strategies that combine multiple models or modalities have been investigated to exploit the complementary strengths of different data sources and improve robustness [21, 2]. However, existing hybrid approaches continue to suffer from several limitations. Many methods still rely on naive strategies such as feature concatenation or fixed-weight fusion, leading to feature redundancy and inefficient integration of meaningful information [25, 7, 32]. Simplistic fusion techniques can inadvertently amplify noise from less informative modalities, particularly in complex or noisy scenes [36]. Cross-modal misalignment remains another major hurdle, as differences in spatial and temporal resolutions between RGB frames, optical flow, and audio streams make it difficult to synchronize modalities effectively [12, 38]. Furthermore, while attention-based fusion techniques have been introduced to dynamically weigh modalities, current designs are often limited in adaptability and may fail to prioritize the most informative modality depending on the anomaly context [18].

Building upon existing research, we adopt a structured multimodal fusion approach that integrates RGB, optical flow, pose, and audio streams. Instead of performing naive concatenation, we apply a semantic-level alignment of multimodal representations before fusion, aiming to enhance consistency across modalities. By employing modality-specific alignment, discrepancies between feature distributions are mitigated, allowing for a more cohesive and informative representation. An adaptive fusion mechanism is also employed to dynamically adjust the contribution of each modality based on contextual importance, which is expected to improve the robustness of anomaly detection.

Specifically, high-level semantic features are extracted using pre-trained models. RGB and optical flow features are obtained through the Inflated 3D ConvNet (I3D) [5], while audio features are extracted using VGGish [15]. To address cross-modal discrepancies, we apply a Modality-wise Feature Matching Subspace (MFMS), which projects features from different modalities into a shared latent space. A contrastive loss function is further incorporated to enforce semantic alignment, encouraging temporally correlated features to be mapped closer in the embedding space. Finally, multimodal features are fused through a Gated Cross-Modal Attention mechanism, allowing the model to adaptively modulate the influence of each modality and focus on the most relevant cues for anomaly detection.

2 Related Works

AED has been studied extensively with the objective of recognizing suspicious activities in video streams. Traditional methods relied on handcrafted features extracted from RGB frames, optical flow, or trajectories [43, 27], but often struggled with the complexity and high dimensionality of real-world video data. Deep learning approaches, particularly convolutional and recurrent neural networks [14, 23], improved the ability to learn spatiotemporal patterns automatically, although challenges in leveraging multiple modalities effectively remained prominent.

Recent research highlights the importance of incorporating optical flow to capture dynamic motion cues that RGB frames alone might miss. Inflated 3D ConvNets (I3D) [5] extended 2D convolutions to 3D, providing strong baselines for spatio-temporal modeling. Work such as RTFM [34] and WSAL [3] demonstrated the effectiveness of using I3D features for weakly supervised AED, achieving high precision in benchmarks such as UCF-Crime and ShanghaiTech. These models emphasized that motion information from optical flow is critical for detecting anomalous patterns, especially in violent or fast-changing scenes.

Complementing visual features, the audio modality introduces an additional layer of context for anomaly detection. Sounds such as alarms, crashes, or screams can signal abnormal events even when visual evidence is weak. The VGGish model [15], pre-trained on large-scale audio datasets, has been widely adopted for extracting semantically meaningful acoustic features. Jin et al. [16] leveraged both audio and visual modalities by integrating cross-modal alignment modules, thereby enhancing detection performance particularly in scenarios with occlusions or noisy environments where single modalities may fall short.

Pose information has emerged as another valuable modality for AED. Instead of relying solely on pixel changes, pose-based methods model human skeletal dynamics, offering a high-level abstraction of actions. Early pose estimation models like OpenPose [4] and lightweight frameworks like Mediapipe [29] provided feasible solutions, but faced challenges with occlusions and temporal coherence. Advanced systems such as PoseC3D [11] leveraged spatiotemporal modeling of 2D joint sequences, achieving strong results in action recognition tasks across datasets like NTU60 and FineGYM. Some recent works integrated pose information into multimodal pipelines to enhance robustness in human-centered anomaly detection. In our context, we adopt RTMDet [24] for human detection, RTMO [22] for keypoint estimation, and PoseC3D for feature extraction, balancing efficiency and accuracy.

Multimodal fusion strategies have been pivotal in advancing AED. Early fusion methods combined features via simple concatenation or static weighting [25, 7], but often led to redundancy or noise amplification. To address these limitations, attention-based dynamic fusion mechanisms were proposed [8, 32], enabling the model to adaptively prioritize modalities depending on the scene context. In particular, Demertzis et al. [8] introduced cross-modal dynamic attention for better feature selection, while Sun et al. [32] focused on multiscale fusion of video and audio streams. Nevertheless, challenges related to cross-modal alignment remained prevalent. Works like Ghadiya et al. [12] and Wei et al. [38] pointed out the difficulties caused by inconsistent feature distributions and asynchronous signals across modalities. Inspired by these insights, our method leverages Modality-wise Feature Matching Subspaces (MFMS) to achieve better semantic alignment before fusion and employs Gated Cross-Modal Attention to adaptively integrate multimodal information.

In parallel, self-supervised learning has emerged as a powerful paradigm to alleviate the dependence on dense annotations. Techniques such as Video Masked Autoencoders (VideoMAE) [35] demonstrated that reconstructing masked video patches enables learning rich spatiotemporal representations without labeled data. The success of self-supervised methods has motivated further research into their integration within multimodal AED pipelines, particularly for underrepresented modalities like audio and pose, enhancing the generalization and robustness of anomaly detection systems in diverse environments.

Building upon these foundations, our proposed method integrates RGB, optical flow, audio, and pose streams within a unified framework. By addressing the challenges of semantic misalignment, redundancy, and dynamic modality importance, we develop a structured fusion and alignment strategy tailored for robust ADE. The detailed design of our approach is described in the methods.

3 Methods

Our approach begins with a multimodal anomaly detection framework designed to capture the complexity of real-world video data. This framework integrates four distinct information streams: RGB, optical flow, audio, and pose. The foundation of our model is inspired by prior research from Jin et al. [16], where the RGB stream captures spatial appearance from raw video frames, the optical flow stream models temporal motion between frames, and the audio stream encodes environmental sound cues. To deepen the model's understanding of human actions, we extend the architecture with a fourth stream dedicated to pose estimation. This stream focuses on skeletal and posture dynamics, providing high-level structural insights into human behavior especially under conditions like occlusion, poor lighting, or background clutter, where visual features may falter.

Instead of aggregating raw features directly, each stream passes through a specialized encoder. The extracted semantic features are aligned in a shared latent space using a Modality-wise Feature Matching

Subspace (MFMS), which reduces inter-modal discrepancies. A GMF mechanism then dynamically balances the contribution of each stream, ensuring that the integrated representation highlights the most relevant information. To support stable training and preserve informative signals, we incorporate a Residual Fusion Block with skip connections.

Learning is guided by a Contrastive Loss that enhances the distinction between normal and anomalous patterns. This is complemented by a Dynamic Loss Weighting strategy, allowing the model to emphasize reliable modalities while suppressing noisy inputs. Optimization is conducted using a Cosine Annealing Learning Rate schedule with warm-up to promote smooth convergence.

For prediction, a regression module replaces the traditional classification head, enabling the model to generate fine-grained anomaly scores. An illustration of our proposed framework is provided in Figure 1, and the subsequent sections offer a detailed breakdown of its components.

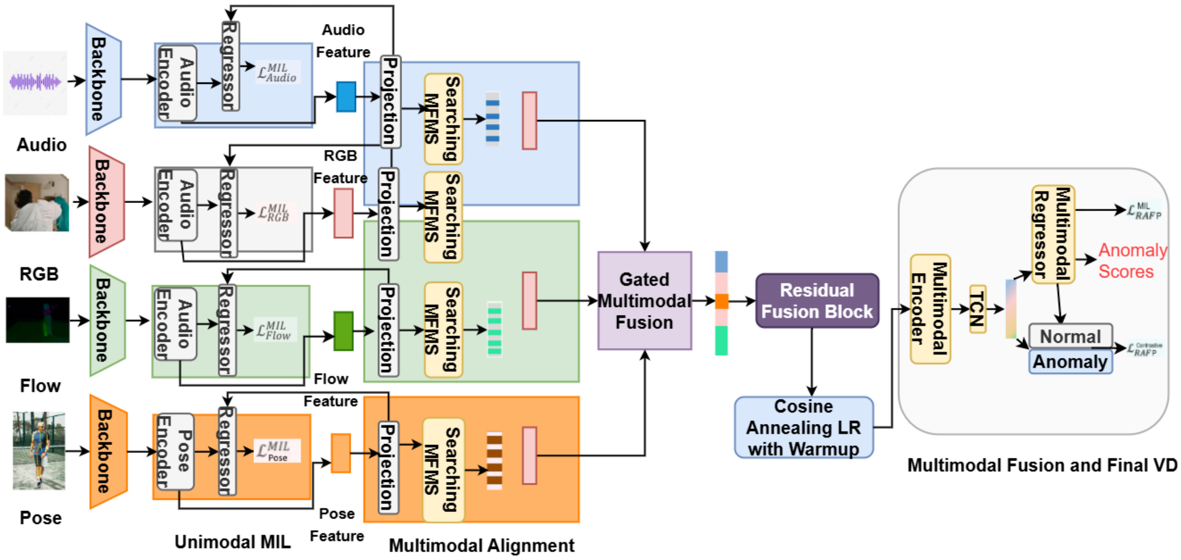

Figure 1: Overview of the Multimodal Anomaly Detection Framework

3.1 Audio Feature Extraction

Audio signals in videos often carry subtle yet vital information for identifying abnormal events, particularly those involving violence. These events frequently exhibit abrupt changes in energy and spectral characteristics, such as gunshots, explosions, or distress calls. In our four-stream fusion framework, the audio modality is responsible for capturing these acoustic cues. Each video is first segmented into non-overlapping chunks of 0.96 seconds. These segments are converted into log-Mel spectrograms, which effectively represent the time-frequency structure of the signal in a perceptually meaningful form.

The spectrograms are then passed through the VGGish model [15], a convolutional neural network pre-trained on the YouTube-100M dataset. This model transforms the audio inputs into 128-dimensional semantic embeddings, forming a temporal feature sequence $F_A \in R^{T\times 128}$, where T is the number of audio segments. and F_A represents the high-level audio feature map.

Although VGGish embeddings effectively capture semantic information, they lack sensitivity to fine-grained temporal variations typical in violent or abnormal sounds. To address this limitation, we apply a temporal refinement module consisting of one-dimensional convolutional layers with exponentially increasing dilation rates, inspired by the architecture in [40]. This setup enables the model to enlarge the receptive field while maintaining the temporal resolution.

The refined output, denoted as F'_A, represents the temporally enhanced audio features and is computed as:

$$F'_A = \text{ReLU}\left(\sum_{i=1}^{N} W_c^{(i)} *_d F_A^{(i-1)} + b_c^{(i)}\right) \tag{1}$$

Here, N is the number of convolutional layers, and $W_c^{(i)}$, $b_c^{(i)}$ are the learnable weights and biases at the i-th layer. The operator $*_d$ denotes dilated convolution with rate $d = 2^{i-1}$. The input to the first layer is defined as $F_A^{(0)} = F_A$, corresponding to the original feature sequence from the VGGish model.

This architecture progressively increases the receptive field while preserving the original temporal granularity of the audio stream. As a result, it enables the detection of both short-term transients, such as gunfire, and longer-duration acoustic patterns, such as screams. Moreover, this design exhibits robustness against stationary background noise due to multi-scale temporal averaging. The fusion of VGGish embeddings and dilated temporal refinement ultimately produces audio features that are both semantically rich and temporally precise, offering critical support for downstream violence detection in complex video environments.

Figure 2 illustrates the entire audio processing pipeline. Raw audio is first segmented into overlapping chunks and converted into log-Mel spectrograms, which are then passed through the VGGish model for feature extraction. The resulting embeddings are further processed through a sequence of dilated 1D convolutional layers followed by normalization, pooling, and fully connected layers for temporal classification.

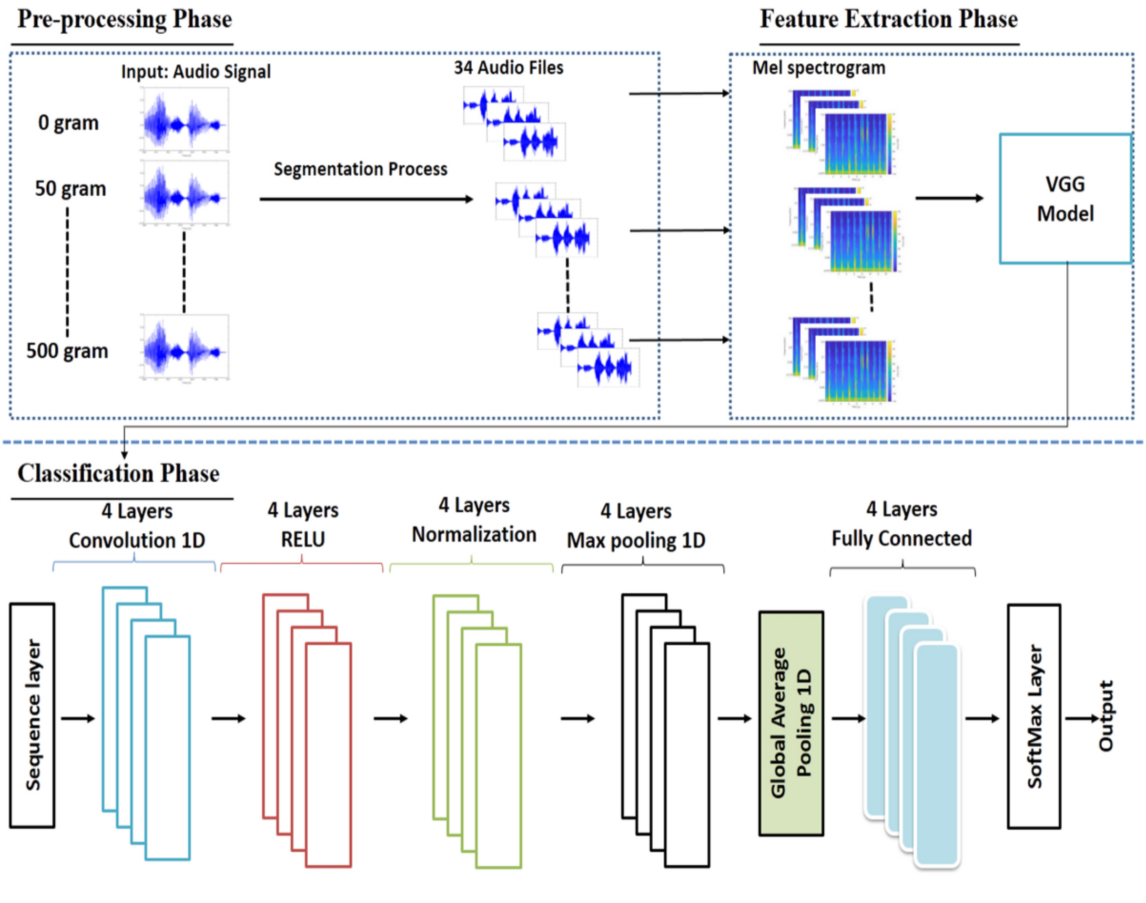

Figure 2: Overview of the audio feature extraction and classification pipeline [19].

3.2 RGB and Flow Feature Extraction

Visual anomalies in videos often manifest through both spatial and motion cues. While RGB frames encode the appearance of objects and their surroundings, optical flow captures the temporal displacement of pixels, reflecting movement between frames. Exploiting the synergy between these two modalities is essential for identifying complex violent behaviors that unfold over time. In our framework, the RGB and optical flow streams are processed in parallel to capture both static and dynamic characteristics of the scene.

To extract spatiotemporal features, we employ the Inflated 3D Convolutional Network (I3D) [5], which extends standard 2D convolutions into the temporal dimension, enabling the network to learn from both appearance and motion across frames. The input to the I3D model comprises RGB frame sequences $X_R \in R^{T\times H\times W\times 3}$ and optical flow sequences $X_F \in R^{T\times H\times W\times 2}$, where T is the number of frames, and $H \times W$ denote spatial resolution. These inputs are processed to produce two separate feature sequences, F_R and F_F, encoding rich spatial and temporal representations.

Despite I3D's capacity to model local spatiotemporal dependencies, violent actions often span longer durations and may involve subtle motion patterns that require extended temporal context. To address this limitation, we refine the output features F_R and F_F by applying a cascade of 1D convolutional layers with exponentially dilated kernels, inspired by [40]. This technique enhances the temporal receptive field without downsampling, thereby preserving fine-grained motion while integrating long-term dependencies.

The refinement process is applied as follows:

$$F'_R = \text{ReLU}\left(W_c^{(2)} *_2 \left(W_c^{(1)} *_1 F_R + b_c^{(1)}\right) + b_c^{(2)}\right) \tag{2}$$

$$F'_F = \text{ReLU}\left(W_c^{(2)} *_2 \left(W_c^{(1)} *_1 F_F + b_c^{(1)}\right) + b_c^{(2)}\right) \tag{3}$$

In these formulations, $W_c^{(i)}$ are 1D convolution kernels with increasing dilation rates $d = 2^{i-1}$, $b_c^{(i)}$ are learnable biases, and $*_d$ denotes the dilated convolution operation. This architecture allows the model to simultaneously attend to short-term and long-term motion cues, such as the sudden onset of aggression or the gradual buildup of abnormal behavior.

The use of dilated convolutions offers two practical advantages: improved robustness to varying motion scales, and resilience to irrelevant background noise. For instance, it enables the model to effectively capture violent activities involving rapid displacement ($\|\Delta xy\|_2 > 5$ pixels/frame) or prolonged pursuits lasting over 50 frames, which are characteristics observed in many AED benchmarks such as XD-Violence [39]. Additionally, applying Gaussian-weighted convolution kernels helps suppress flow estimation artifacts, ensuring the network focuses on motion saliency.

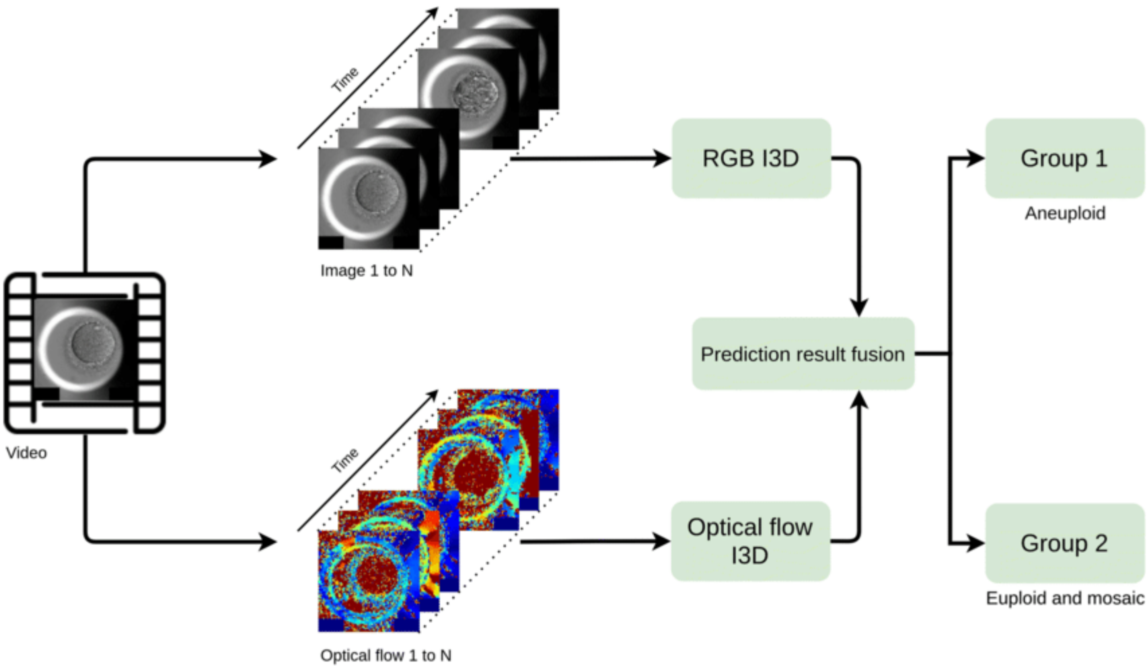

Figure 3: Flow feature processing pipeline [20]

Combining I3D features with dilated refinement enables our model to capture diverse motion patterns while preserving spatial coherence. This robust RGB-flow integration strengthens anomaly detection under occlusion, clutter, and scene dynamics, forming a key foundation of our system.

3.3 Pose Feature Extraction

While RGB and optical flow streams capture valuable appearance and motion cues, they exhibit notable limitations in scenarios involving low illumination, occlusions, or cluttered backgrounds. These visual modalities may also fail to differentiate subtle but semantically distinct human actions due to their reliance on low-level pixel information. To overcome these constraints, we incorporate a dedicated pose stream that models human skeletal dynamics, offering a higher-level and more interpretable representation of body movements.

This additional modality provides crucial kinematic insights, focusing on the configuration and temporal evolution of joints rather than raw pixel changes. Such abstraction enables the system to distinguish between visually similar yet contextually different actions, such as falling versus picking up an object. Moreover, pose features are inherently invariant to appearance changes such as clothing, lighting, and background, thereby improving the robustness and generalization of the model across diverse environments.

During the development of the pose-based behavior recognition pipeline, several pose estimation models were evaluated before finalizing the selection. Alternatives such as OpenPose, MediaPipe Pose Landmarker, MMPose, PoseTrack21, RTMLib, and SAPIENS were considered. OpenPose produced accurate results but suffered from slow processing, limiting its suitability for real-time applications. MediaPipe Pose Landmarker offered efficient performance on edge devices but exhibited reduced accuracy under crowded or occluded scenarios. MMPose achieved high accuracy and flexible backbone support, yet its complexity posed deployment challenges. PoseTrack21 emphasized pose tracking across frames but did not align well with motion-specific feature modeling. SAPIENS provided semantically rich representations but required complicated configurations and sometimes unavailable 3D annotations. After extensive testing, the combination of RTMDet for human detection, RTMO for keypoint detection, and PoseC3D for spatiotemporal modeling was selected for its balanced trade-off between speed, accuracy, and stability, ensuring consistent performance across varying video conditions.

We utilize the PoseC3D framework [10], which processes 2D joint coordinates and transforms them into a 3D spatiotemporal representation for skeleton-based action recognition. The pipeline consists of three key stages. First, RTMDet is employed for human detection, providing stable person localization with a favorable balance between computational efficiency and detection accuracy. Second, the detected bounding boxes are passed to RTMO, a fast one-stage keypoint estimator that yields per-frame joint coordinates in the format (x, y, c), where c denotes the confidence score.

Once keypoints are obtained, each pose is converted into a joint heatmap of shape $K \times H \times W$ by generating 2D Gaussian maps centered at each joint. Each joint heatmap $J_k(i, j)$ represents the spatial confidence of joint k being at position (i, j), defined as:

$$J_k(i,j) = e^{-\frac{(i-x_k)^2+(j-y_k)^2}{2\sigma^2}} c_k \tag{4}$$

where (x_k, y_k) and c_k are the coordinates and confidence score of the k-th joint. Additional limb heatmaps are constructed by computing the distance from each spatial point to the line segment connecting joint pairs. Each limb heatmap $L_k(i, j)$ is given by:

$$L_k(i,j) = e^{-\frac{D((i,j),\text{seg}[a_k,b_k])^2}{2\sigma^2}} \min(c_{a_k}, c_{b_k}) \tag{5}$$

These joint and limb heatmaps are stacked along the temporal dimension to form a 3D volume of size $K \times T \times H \times W$, encoding both spatial and temporal joint behavior.

Rather than processing entire frames, subject-centered cropping is applied to localize the smallest bounding region encompassing all joints across frames, thereby enhancing processing efficiency while retaining critical motion information. The cropped sequences are resized to a fixed spatial resolution. A uniform temporal sampling strategy ensures that representative frames are selected across the video's duration, preserving the global temporal structure of human actions.

For feature aggregation, Global Average Pooling (GAP) is applied to compress spatial dimensions. At each time step t, a spatial feature vector F_t is computed as:

$$F_t = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} V(i,j,t) \tag{6}$$

where $V(i, j, t)$ denotes the heatmap value at spatial location (i, j) and time step t. To obtain a fixed-length temporal sequence across videos, bilinear interpolation is used to normalize the number of frames. The resulting temporally aligned feature vector F'_t is defined as:

$$F'_t = \sum_{t=1}^{T} F_t \cdot \max(0, 1 - |t - t'|) \tag{7}$$

This produces a temporally aligned feature sequence F' of fixed length, which is subsequently passed to the multimodal fusion module.

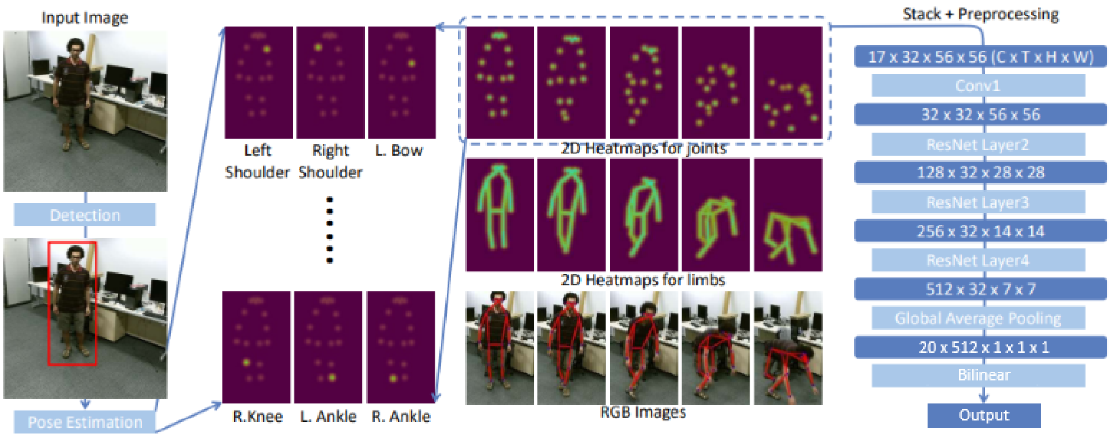

Figure 4: Pose feature processing pipeline.

The addition of pose features significantly enhances the model's ability to recognize anomalous behaviors, particularly those that may remain visually ambiguous when relying solely on appearance or motion cues. In complex surveillance environments, skeletal motion patterns extracted via this pipeline offer clearer indicators of human intent and interaction dynamics.

3.4 Unified Multi-Modal Anomaly Detection Framework

The proposed framework integrates four complementary information streams consisting of RGB, optical flow, audio, and pose into a structured processing pipeline for weakly supervised anomaly detection. The processing begins with per-modality encoding to extract high-level semantic features, followed by a semantic alignment stage and a gated fusion mechanism. A block diagram of the overall architecture is shown in Figure 1.

Instead of relying on early fusion or direct concatenation of modality features, our framework introduces a structured two-stage approach: semantic alignment followed by multimodal fusion. Each modality RGB, flow, audio, and pose is first processed by a dedicated unimodal encoder consisting of a temporal convolutional layer and a Transformer-based self-attention module. These encoders extract high-level semantic embeddings specific to each stream. The pose features are derived from PoseC3D using 2D joint keypoints and limb heatmaps, then converted into spatiotemporal volumes to capture motion dynamics.

To reduce modality discrepancy and redundancy, we align secondary modalities (flow, audio, pose) into the semantic space of the primary modality (RGB) through projection layers. A sparse alignment mechanism is applied, where modality-specific embeddings are projected into their most informative subspaces via MFMS. This alignment is guided by cosine similarity and constrained using both embedding-level and score-level losses, ensuring semantic and temporal consistency across streams.

Following alignment, modality features are concatenated and fused using a GMF module, which dynamically adjusts the contribution of each modality. A Residual Fusion Block further enhances this representation by reinforcing original modality signals and stabilizing gradients. The fused representation is passed through a TCN to model higher-level temporal dependencies, yielding frame-wise anomaly scores.

The entire system is trained using a composite loss function that includes unimodal MIL loss, alignment loss, multimodal MIL loss, and a triplet-based contrastive loss to improve separation between normal and violent segments. Ground truth labels are provided only at the video level, making the training procedure fully weakly supervised. Despite this, our framework demonstrates strong performance and generalization capability, particularly in scenes with occlusion, low lighting, or subtle behavioral anomalies.

Experimental results validate that the inclusion of the pose stream and the semantic alignment strategy offer tangible benefits over traditional concatenation-based fusion approaches. Our ablation studies further confirm that each modality contributes uniquely, and that semantic alignment significantly enhances the effectiveness of fusion. The four-modal design is thus not a redundant extension, but a critical advancement toward building robust and generalizable video anomaly detection systems.

3.5 Multimodal Alignment

Following the extraction of modality-specific features from RGB, optical flow, audio, and pose streams, the next critical step involves reconciling their semantic and temporal discrepancies before conducting effective fusion. As discussed previously, these modalities capture complementary aspects of abnormal events, yet inherently differ in their statistical properties, temporal granularity, and semantic abstraction levels. Directly fusing unaligned features could lead to representational inconsistencies, redundant information, or noise amplification, ultimately degrading anomaly detection performance.

The MFMS module is introduced as a structured projection mechanism to align modality-specific features into a shared latent space. Unlike naive feature concatenation, MFMS learns projection functions for each modality that map their respective embeddings into a semantically consistent space centered around the RGB stream. This process mitigates modality discrepancies while preserving the distinctive discriminative characteristics of each modality, ensuring coherent feature interactions during fusion.

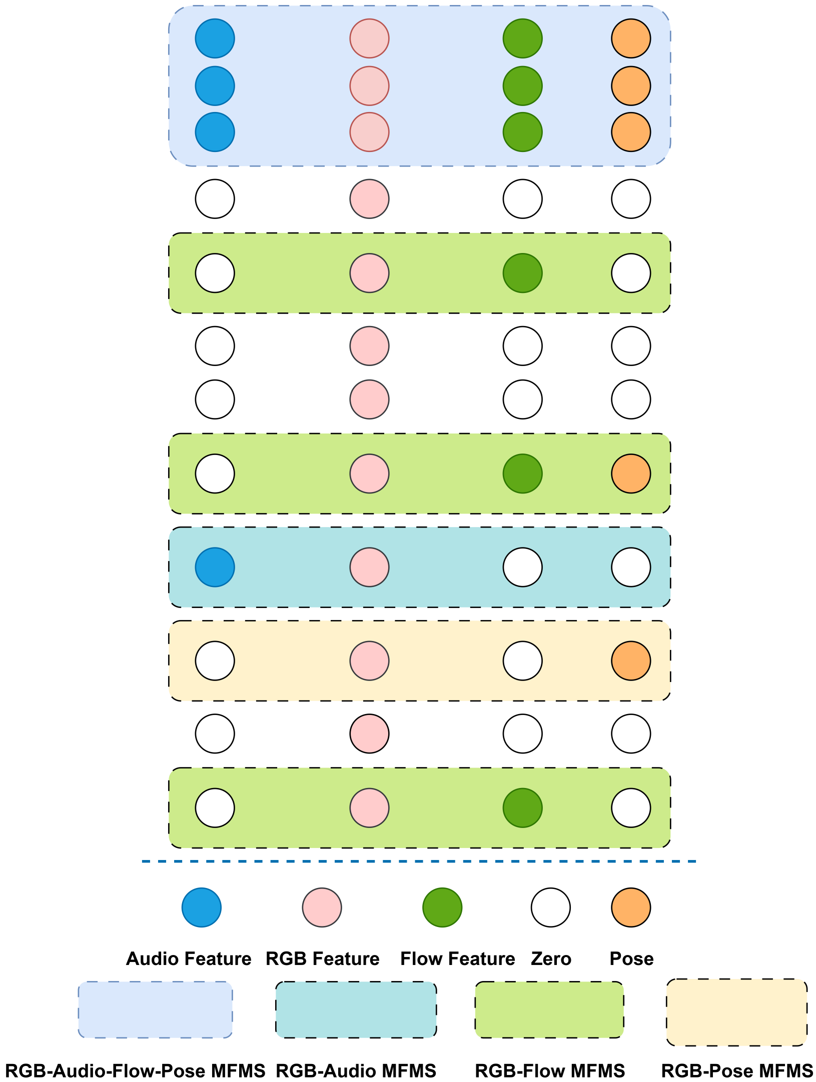

Figure 5: MFMS for modality alignment. The process includes linear projection of each modality's features into a shared latent space followed by intra-modality refinement using self-attention and inter-modality interaction using cross-attention.

The attention-augmented feature representations A_m extracted from each modality $m \in \{\text{RGB, Flow, Audio, Pose}\}$ are linearly transformed as:

$$Z_m = W_m A_m + b_m \tag{8}$$

where $W_m \in R^{d' \times d_m}$ and $b_m \in R^{d'}$ are learnable parameters projecting features into a unified latent space $R^{d'}$. This transformation standardizes feature distributions while retaining key modality-specific attributes necessary for downstream tasks.

An intra-modality refinement is then applied through self-attention mechanisms, enabling temporal consistency enhancement and selective emphasis on salient information within each modality:

$$\tilde{Z}_m = \text{softmax}\left(\frac{Q_m K_m^T}{\sqrt{d'}}\right) V_m \quad \text{where} \quad \begin{cases} Q_m = W_{q,m} Z_m \\ K_m = W_{k,m} Z_m \\ V_m = W_{v,m} Z_m \end{cases} \tag{9}$$

where $W_{q,m}, W_{k,m}, W_{v,m} \in R^{d' \times d'}$ are modality-specific projection matrices.

Cross-modal dependencies are modeled through an inter-modality interaction step, where features from different modalities interact via cross-attention:

$$Z'_m = \text{softmax}\left(\frac{Q_m K_n^T}{\sqrt{d'}}\right) V_n \quad \text{where} \quad \begin{cases} Q_m = W_{q,m} \tilde{Z}_m \\ K_n = W_{k,n} \tilde{Z}_n \\ V_n = W_{v,n} \tilde{Z}_n \end{cases} \quad \text{for} \quad m \neq n \tag{10}$$

This cross-modal refinement enables each modality to incorporate contextually relevant information from the others while preserving its core characteristics.

Alignment quality is further reinforced by minimizing a feature distributional divergence loss across modalities:

$$\mathcal{L}_{\text{align}} = \sum_{m \neq n} \|\phi(Z'_m) - \phi(Z'_n)\|_2^2 \tag{11}$$

where $\phi(\cdot)$ denotes feature statistics such as mean and variance.

By systematically integrating modality projection, intra-modality refinement, and inter-modality interaction into a unified framework, the MFMS module produces rich, structured, and semantically aligned multimodal representations. This alignment stage establishes a coherent foundation for the subsequent fusion process, ultimately enhancing the model's capability in detecting anomalies in complex real-world video data.

3.6 Gated Multimodal Fusion and Anomaly Detection

An effective anomaly detection system must not only align multimodal features but also integrate them in a context-aware manner. While the MFMS module ensures semantic coherence across modalities, the importance of each modality still varies depending on scene conditions. A uniform fusion approach may amplify noise or suppress critical cues, such as subtle skeletal movements from pose data or motion artifacts in optical flow.

The GMF module dynamically modulates the contribution of each modality by applying a learned gating mechanism to the aligned features $Z'_{\text{RGB}}, Z'_{\text{Flow}}, Z'_{\text{Audio}}, Z'_{\text{Pose}}$. Each stream is reweighted based on learned attention using:

$$G_m = \sigma(W_{g,m} Z'_m + b_{g,m}), \quad \forall m \in \{\text{RGB}, \text{Flow}, \text{Audio}, \text{Pose}\} \tag{12}$$

where $W_{g,m} \in R^{d' \times d'}$ and $b_{g,m} \in R^{d'}$ are learnable parameters, and $\sigma(\cdot)$ is the sigmoid activation. The final fused representation is computed as:

$$F_{\text{fused}} = \sum_m G_m \odot Z'_m \tag{13}$$

with $\odot$ denoting element-wise multiplication. This selective fusion enables the model to prioritize informative signals while suppressing irrelevant or noisy inputs, as demonstrated in similar multimodal attention architectures [8, 32].

A Residual Fusion Block is then employed to enhance stability and retain original feature richness. The fused output undergoes a lightweight transformation via a linear layer and is normalized using LayerNorm before being added back through a skip connection:

$$F_{\text{res}} = \text{LayerNorm}(F_{\text{fused}} + W_r F_{\text{fused}}) \tag{14}$$

where $W_r \in R^{d' \times d'}$ is a trainable projection. This structure mitigates vanishing gradients and ensures that fine-grained features such as body posture or subtle acoustic transitions are preserved.

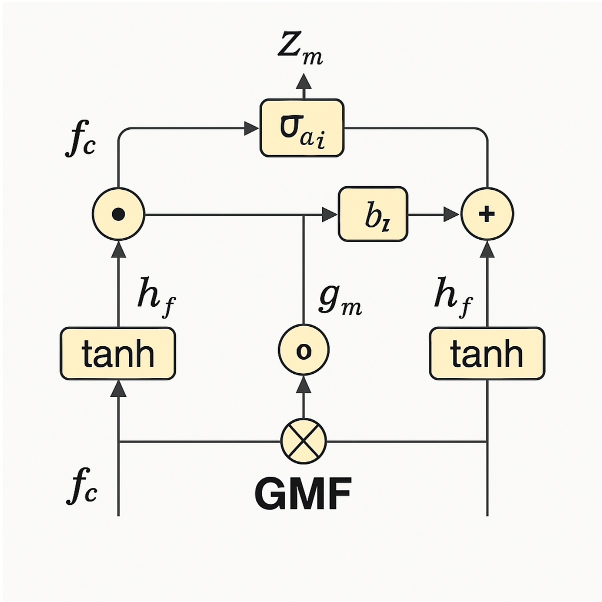

Figure 6: Illustration of the GMF process. Each modality feature f_c is first processed through a gating mechanism to compute the gating factor g_m, which determines its contextual importance. The original feature is then reweighted by the gating factor, and all reweighted modalities are aggregated to form the fused representation.

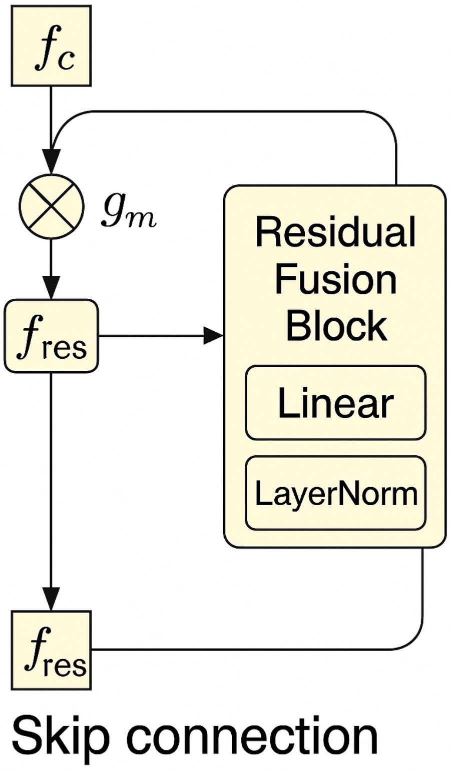

Figure 7: Residual Fusion Block incorporating skip connection and LayerNorm to stabilize and enrich the fused representation.

After fusion and refinement, the representation F_{res} is passed to a TCN, which captures both short-term and long-range dependencies via dilated convolutions. This temporal modeling is crucial for recognizing evolving abnormal patterns.

An MIL-based scoring head subsequently maps the temporally contextualized features to frame-level anomaly scores:

$$S_t = W_s\left(\mathrm{ReLU}(W_c F_{\text{res}} + b_c)\right) + b_s \tag{15}$$

where $W_c \in R^{d' \times d'}$, $b_c \in R^{d'}$, $W_s \in R^{1 \times d'}$, and $b_s \in R$ are trainable parameters. The output S_t is normalized to [0,1] via min-max scaling for consistency across sequences.

The combined effect of gated fusion, residual propagation, and temporal modeling results in a potent multimodal framework capable of detecting both overt and subtle anomalies in real-world surveillance videos.

3.7 Loss Formulation and Contrastive Loss

Following gated fusion and temporal modeling, the optimization process is guided by a structured combination of loss functions designed to balance learning objectives across modalities and maintain potent feature discrimination. The total loss is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{UMIL}} + \mathcal{L}_{\text{MAL}} + \mathcal{L}_{\text{MMIL}} + \mathcal{L}_{\text{Contrastive}} \tag{16}$$

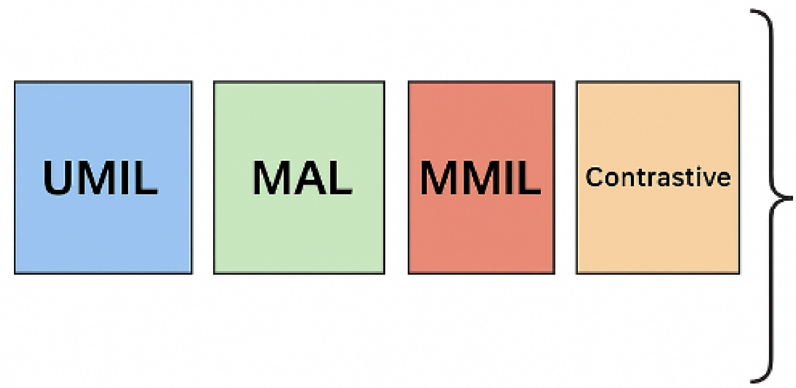

Figure 8: Structured combination of loss functions for potent optimization.

The Unimodal MIL Loss ($\mathcal{L}_{\text{UMIL}}$) supervises the learning of discriminative features within each modality separately, ensuring that even without multimodal interaction, strong unimodal representations are maintained. The Modality Alignment Loss ($\mathcal{L}_{\text{MAL}}$) enforces consistency between the projected embeddings of different modalities, thereby supporting semantic coherence throughout the fusion pipeline. The Multimodal MIL Loss ($\mathcal{L}_{\text{MMIL}}$) supervises anomaly detection based on the fused multimodal features, encouraging the integrated representation to capture complex cross-modal dependencies indicative of abnormal events. The Contrastive Loss ($\mathcal{L}_{\text{Contrastive}}$) further refines the feature space by pulling temporally or semantically similar segments closer while pushing dissimilar ones apart, reinforcing discriminability in the aligned space.

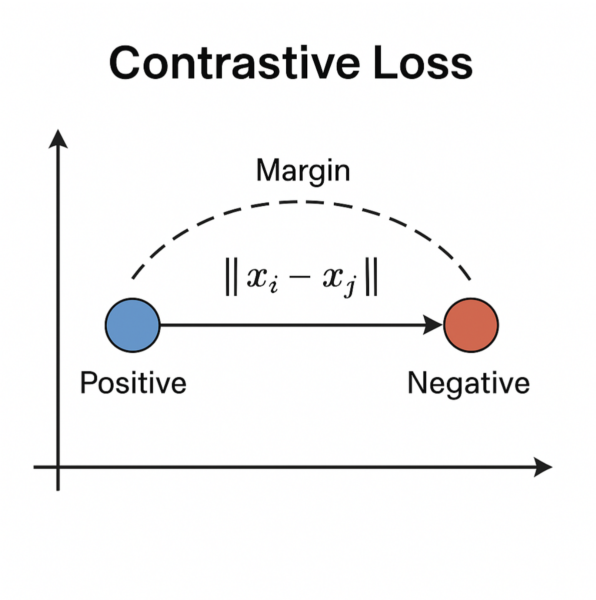

Figure 9: Contrastive loss mechanism for enhancing feature discriminability.

Gradient stability is maintained during training through gradient clipping, preventing the magnitude of updates from exceeding a predefined threshold:

$$\nabla_\theta \mathcal{L}_{\text{total}} = \text{clip}\left(\nabla_\theta \mathcal{L}_{\text{total}}, |\nabla|_2 \leq \gamma\right) \tag{17}$$

Here, γ is a tunable hyperparameter that controls the maximum allowed gradient norm. In our implementation, we set $\gamma = 1.0$, a standard choice widely adopted in deep learning to prevent exploding gradients while ensuring stable convergence [42]. This threshold is listed in Table 1 under "Gradient Clipping Threshold". The setting allows for stable optimization without impeding gradient flow during training.

By jointly optimizing these losses, the model benefits from strong unimodal learning, consistent cross-modal alignment, potent multimodal fusion, and enhanced separability in the embedding space. This comprehensive loss design ultimately equips the system with the ability to detect both clear and subtle anomalies in diverse and noisy real-world surveillance scenarios.

4 Results And Discussion

4.1 Dataset and Evaluation Metric

The XD-Violence dataset[39], the largest publicly available multimodal dataset for weakly supervised violence detection, is used for evaluation. It consists of 4,754 untrimmed videos (total 217 hours) from diverse sources such as movies, surveillance footage, and YouTube clips. Each video contains RGB frames, optical flow,pose features and audio, allowing for comprehensive multimodal analysis. The dataset is split into 3,954 training videos (1,905 violent, 2,049 non-violent) and 800 testing videos (500 violent, 300 non-violent). This multimodal structure enhances the model's ability to represent violent and abnormal events effectively.

For evaluation, we adopt frame-level Average Precision (AP) as the primary metric, following prior works. AP assesses the model's capability to localize abnormal events at a fine-grained level by computing the area under the precision-recall curve. Formally, it is defined as:

$$\text{AP} = \int_0^1 p(r)\, dr \tag{18}$$

where $p(r)$ is the precision as a function of recall r.

Additionally, we use the Area Under the Curve (AUC) of the precision-recall curve to measure the model's overall ability to discriminate between violent and non-violent frames. It is calculated by integrating the precision-recall function:

$$\text{AUC} = \int_0^1 p(r)\, dr \tag{19}$$

This metric reflects how well the model distinguishes abnormal segments under varying confidence thresholds. Higher AUC values indicate better discrimination performance. Both metrics provide complementary insights: AP focuses on the model's ability to localize event boundaries precisely, while AUC quantifies overall discriminative capability across frames. Finally, we compare our results with state-of-the-art methods to ensure fair benchmarking and highlight performance improvements.

4.2 Implementation Details

We outline the setup and configuration used for evaluating our model. Our model was implemented using PyTorch, and we primarily focus on the configuration of hyperparameters to ensure optimal model performance and follow the standard multimodal fusion pipeline to ensure fair comparisons with existing methods. Our experiments were conducted on a computing system equipped with an Intel i5 12600K CPU, 32GB of RAM, and an Nvidia RTX Geforce 3060 GPU, ensuring the computational capacity to handle the large dataset and model training. To ensure potent training, we used the following hyperparameter settings:

Table 1: Hyperparameter Setup

Parameter	**Value**
Frame Rate	24 FPS
Video Window Size	16 frames
Audio Segment Length	960 ms
Seed	500
Batch Size	64
Optimizer	Adam
Learning Rate	0.0001
Betas (Adam)	(0.9, 0.999)
Weight Decay	0.0005
Scheduler	Cosine Annealing with Warmup
Gradient Clipping Threshold (γ)	1.0
Training Iterations	1,200

4.3 Experimental Results on XD-Violence

Table 2 presents the performance comparison of various methods on the XD-Violence dataset, categorized by their learning paradigms and the number of modalities they utilize. Among unimodal methods relying solely on RGB features, the SVM baseline achieves an AP of 50.78%, while unsupervised methods such as OCSVM [28] and Hasan et al. [14] achieve lower APs of 27.25% and 30.77%, respectively. In contrast, weakly supervised approaches such as Sultani et al. [30] and RTFM [34] show notable improvements, reaching APs of 75.68% and 77.81%, respectively.

When incorporating two modalities (RGB and audio), further performance gains are observed. HL-Net [39] achieves an AP of 78.64%, ACF [37] achieves 80.13%, and Zhang et al. [41] reaches 81.43%. These results reinforce the importance of audio cues for violence detection, especially when visual information alone is insufficient.

Moving to three-modal fusion methods (RGB, audio, and optical flow), MSBT [31] achieves an AP of 84.32%, demonstrating the benefits of leveraging temporal motion cues. Recent stronger baselines such as HyperVD [26] and MAVD [16] achieve even higher APs of 85.67% and 86.07%, respectively. CFA-HLGAtt [13], another strong competitor, achieves 86.34%.

Our proposed model achieves an AP of 83.51% when using RGB, audio, and flow modalities, showing competitive results with existing strong methods. Importantly, by incorporating pose information as an additional modality, the AP improves further to 84.98%. The pose stream captures subtle skeletal motion dynamics that may not be fully represented by appearance or flow alone. This highlights the effectiveness of structural human dynamics for detecting complex abnormal behaviors.

In addition to modality expansion, enhancing the fusion strategy through GMF significantly strengthens the model's capability. By dynamically adjusting the importance of each modality based on contextual relevance, GMF prevents dominant modalities from overwhelming more subtle but informative signals, such as those captured by pose data. Compared to uniform or naive fusion, GMF facilitates better prioritization of critical cues and suppresses noise, contributing meaningfully to the overall performance gains.

Table 2: Performance Comparison on the XD-Violence Dataset

Method	Modalities	AP (%)
SVM baseline	RGB	50.78
OCSVM [28]	RGB	27.25
Hasan et al. [14]	RGB	30.77
Sultani et al. [30]	RGB	75.68
RTFM [34]	RGB	77.81
HL-Net [39]	RGB + Audio	78.64
ACF [37]	RGB + Audio	80.13
Zhang et al. [41]	RGB + Audio	81.43
MSBT [31]	RGB + Audio + Flow	84.32
HyperVD [26]	RGB + Audio	85.67
MAVD [16]	RGB + Audio + Flow	86.07
CFA-HLGAtt [13]	RGB + Audio	86.34
Ours (3 modalities)	RGB + Audio + Flow	83.51
Ours (+Pose)	RGB + Audio + Flow + Pose	84.98

While our method showed a steady increase in performance, we observed stagnation in improvement beyond 850 training iterations. Initially, the multimodal fusion strategy led to significant performance gains, but after 850 epochs, the model's loss values stopped decreasing, and the AP plateaued. Learning rate adjustments and dynamic loss weighting between 800 and 900 iterations helped stabilize training but did not yield further accuracy improvements. Table 3 provides detailed performance metrics at iteration 850.

Table 3: Training Metrics at Iteration 850

Iteration	mAP	Step	U-MIL Loss	MA Loss	M-MIL Loss	Contrastive Loss	Learning Rate
850	84.98%	850	0.925	4.3241	0.0341	0.6939	9.52×10^{-5}

Following these observations, we conducted extensive ablation studies to further evaluate the individual contributions of modality expansion, fusion mechanisms, and loss function design, as discussed in the subsequent sections.

Ablation Studies on Modality Expansion, Fusion Strategy, Loss Design, and Alignment Strategy Ablation studies were conducted to systematically evaluate the contributions of the expanded modality set, the GMF mechanism, the integration of the Contrastive Loss, and the use of the MFMS. The evaluation covers four key aspects: the effect of adding the pose modality, the benefit of adopting the GMF strategy, the impact of replacing the Triplet Loss with a Contrastive Loss formulation, and the importance of performing cross-modal alignment through MFMS.

First, for modality expansion, Table 4 compares the performance between the model using three modalities (RGB + Audio + Flow) and the model incorporating pose information (RGB + Audio + Flow + Pose). Adding the pose stream improves the AP from 83.51% to 84.34%, highlighting that skeletal motion dynamics provide valuable cues for detecting subtle abnormal behaviors that may not be fully captured through RGB and optical flow alone.

Table 4: Impact of Adding Pose Modality

Setting	Modalities	AP (%)
Ours (3 modalities)	RGB + Audio + Flow	83.51
Ours (+Pose)	RGB + Audio + Flow + Pose	84.34

Second, regarding the fusion strategy, Table 5 shows the effect of replacing naive feature concatenation with the Gated Multimodal Fusion mechanism. The GMF approach further boosts the AP by approximately 0.64%, demonstrating that adaptively reweighting modalities based on contextual relevance enhances the system's potency and precision.

Table 5: Impact of Gated Multimodal Fusion

Fusion Strategy	AP (%)
Naive Concatenation	83.70
Gated Multimodal Fusion (GMF)	84.34

Third, Table 6 presents the effect of replacing the Triplet Loss with a Contrastive Loss formulation. This substitution leads to an additional gain from 84.34% to 84.98% AP. Integrating Contrastive Loss helps to enforce stronger semantic consistency across modalities, allowing the model to better cluster similar events while separating abnormal behaviors from normal activities. This result confirms that Contrastive Loss contributes not only to better alignment across multimodal features but also to improved discriminability in complex surveillance scenarios.

Table 6: Impact of Contrastive Loss Integration

Loss Type	AP (%)
Triplet Loss	84.34
Contrastive Loss	84.98

Finally, to quantify the effect of cross-modal alignment, Table 7 shows the impact of removing the MFMS module. Without MFMS, the AP drops from 84.98% to 84.20%, indicating that semantic misalignment between modalities can severely hinder fusion quality. The MFMS projection step thus plays a critical role in enhancing feature coherence across different modalities, ultimately leading to more accurate anomaly detection.

Table 7: Impact of Modality-wise Feature Matching Subspace (MFMS)

Alignment Strategy	AP (%)
Without MFMS	84.20
With MFMS (Ours)	84.98

These results collectively demonstrate that the integration of the pose modality, the adoption of the gated fusion mechanism, the use of Contrastive Loss, and the application of MFMS alignment are key factors contributing to the superior performance achieved by the proposed multimodal anomaly detection framework.

4.4 Qualitative Analysis

We delve deeper into the qualitative aspects of our model's performance by analyzing the training dynamics, loss functions, and accuracy trends, as well as the AUC for AP. These visualizations provide further insights into the model's learning behavior and the effects of our multimodal fusion approach.

Training Loss and AP Trends Figure 10 presents the convergence behavior of the loss functions and the corresponding AP over the course of training. The figure shows the evolution of the four loss components U-MIL Loss, MA Loss, M-MIL Loss, and Contrastive Loss along with AP. As training progresses, we observe that the U-MIL Loss and M-MIL Loss decrease significantly, indicating that the model is successfully learning from the multimodal features. Meanwhile, the MA Loss and the Contrastive Loss stabilize, suggesting that the model has reached a plateau in learning these components. Importantly, the AP (represented on the left axis) improves steadily, reaching around 80% by the end of training, which indicates the model's increasing ability to correctly distinguish violent and non-violent frames. However, after about 1000 iterations, the AP curve shows signs of stagnation, suggesting that the model might benefit from further fine-tuning or adjustments in the fusion strategy.

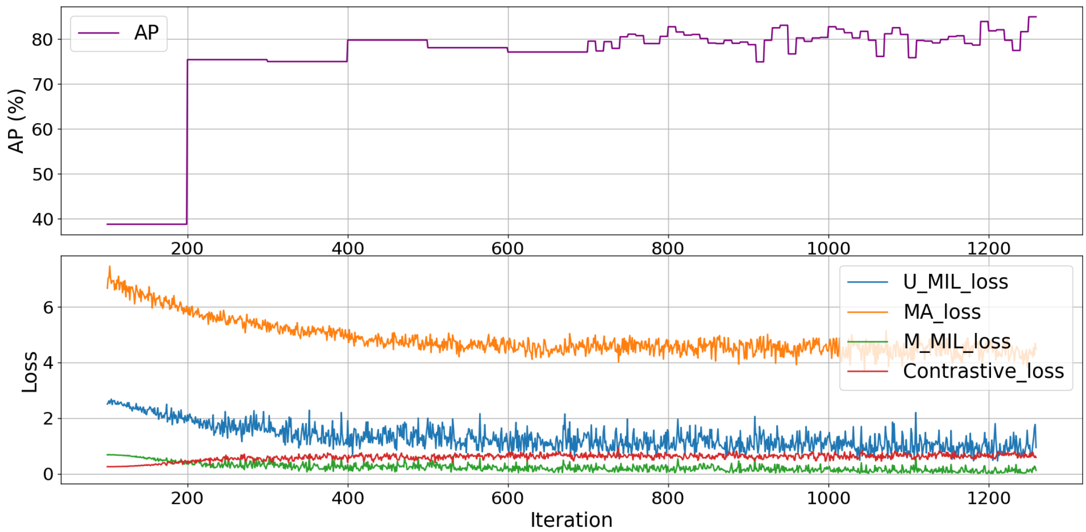

Figure 10: Training Loss and Accuracy Trends of Our Model.

Area Under the Curve (AUC) Figure 11 demonstrates the Area Under the Curve (AUC) for Average Precision (AP) over the iterations. The AUC is a key evaluation metric that quantifies the model's ability to distinguish between violent and non-violent frames. The plot shows a steep increase in AUC in the initial training phases, reaching a value of AUC = 87263.66, which indicates a strong ability to differentiate between the two classes. As the training progresses, the AUC value stabilizes, suggesting that the model has successfully captured the key features necessary for classifying abnormal events. In the context of violence detection, a higher AUC implies that the model is capable of correctly identifying abnormal behaviors with high confidence across varying decision thresholds, reflecting strong generalization and robustness. Conversely, a lower AUC indicates limited discrimination capacity, which may result in missed detections or increased false alarms when the input conditions change. The AUC curve complements the training loss analysis by providing a clear and interpretable measure of the model's discrimination ability. The consistency and plateauing of the AUC curve align with the observed training stagnation, reinforcing the need for possible adjustments to the learning process, such as tuning the learning rate or exploring more advanced fusion techniques.

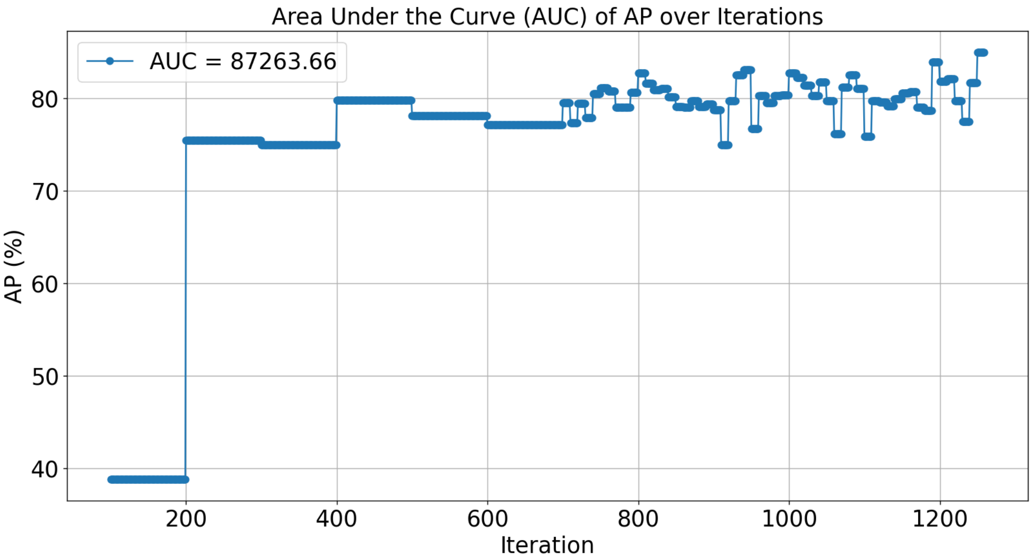

Figure 11: Area Under the Curve (AUC) of AP over Iterations

Insights and Future Directions Throughout the training process, our analysis of the loss dynamics and evaluation metrics such as accuracy and AUC reveals a clear trend: the model demonstrates rapid performance improvements during the early training stages, particularly due to the benefits introduced by structured multimodal fusion and temporal refinement. This validates the effectiveness of our architectural design in capturing complementary information from multiple modalities. However, as training progresses beyond approximately 850 epochs, both loss values and performance metrics begin to plateau, suggesting a saturation point in the learning capacity of the model under the current configuration. This phenomenon may be attributed to two primary factors: first, the potential overfitting arising from prolonged exposure to the training data without sufficient regularization or variation; and second, the limited adaptability of the existing fusion mechanism and loss weighting strategy in later stages of optimization. These observations point to the need for more flexible and responsive training methodologies. As a direction for future work, we plan to investigate advanced multimodal alignment strategies that can dynamically recalibrate intermodal relationships based on contextual relevance. In addition, incorporating adaptive loss weighting mechanisms in which the contribution of each modality to the overall loss is adjusted based on confidence or informativeness could improve the model resilience to noisy or incomplete input. We also intend to explore alternative learning rate scheduling techniques, such as cosine annealing with warmup or cyclical learning rates, which may facilitate smoother convergence and mitigate stagnation in the optimization process. Finally, we are interested in experimenting with contrastive or hybrid loss functions that encourage better separation between normal and anomalous patterns in the feature space. Together, these future directions aim to enhance the robustness, generalizability, and practical utility of our framework for real-world multimodal anomaly detection applications.

Generalization Test on Out-of-Distribution Videos Building upon the observations from the XD-Violence dataset, we further evaluated the generalization capability of our model by conducting inference on two out-of-distribution (OOD) videos. These videos were not included in the training set and represent scenarios with substantial domain shifts. The first video captures a UFC fight, showcasing intense physical confrontations, while the second video depicts a non-violent romantic interaction.

The anomaly score timeline for the UFC fight video is illustrated in Figure 12. Throughout the match, the model successfully detected major violent activities, as indicated by sharp peaks in the anomaly score curve aligning with physical strikes, knockdowns, and ground engagements. During non-violent intervals such as referee interventions or post-fight ceremonies, the anomaly scores remained relatively low, demonstrating the model's capability to modulate its predictions based on contextual cues. Quantitative evaluation further supports these findings, with a best F1-score of 46.09

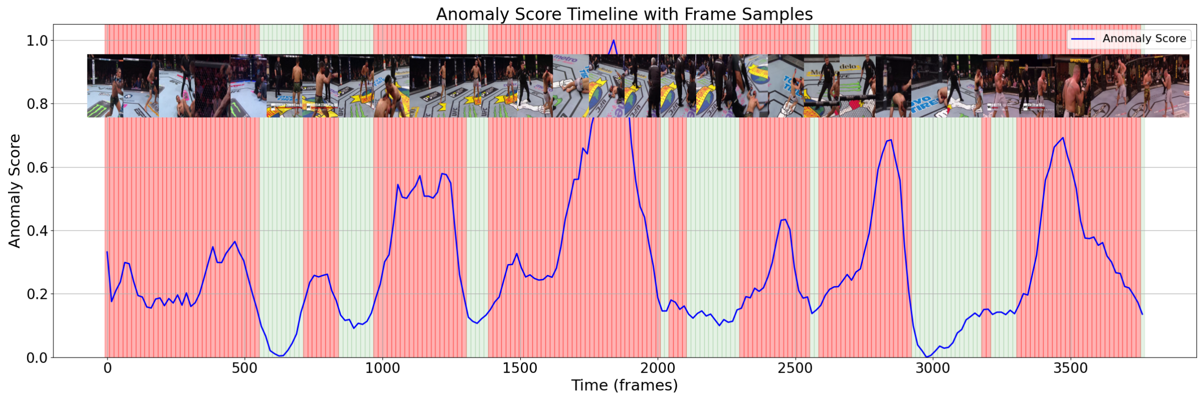

Figure 12: Anomaly score timeline on an out-of-distribution UFC fight video. Pink areas mark manually labeled violent events.

In contrast, the inference result on a romantic scene video, shown in Figure 13, reveals a stable prediction pattern with very low anomaly scores across frames. Although minor fluctuations occurred during moments of dynamic gestures such as hugging, the model refrained from erroneously classifying these interactions as violent. The corresponding evaluation metrics confirm this qualitative observation, with an overall accuracy of 83.33% and both precision and recall remaining at 0%, indicating the absence of false alarms. This outcome emphasizes the model's robustness in distinguishing aggressive behaviors from benign human activities, maintaining a low false positive rate even in unfamiliar, non-violent contexts.

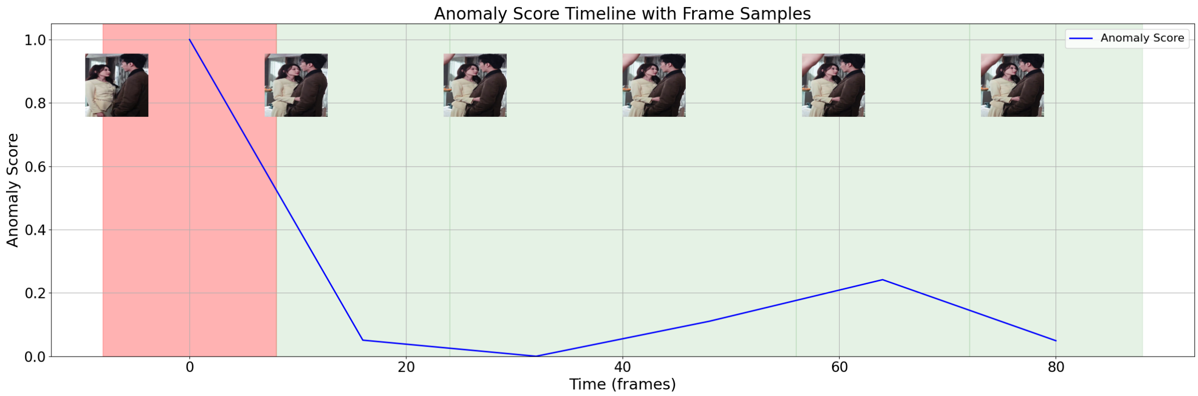

Figure 13: Anomaly score timeline on a non-violent romantic scene video.

These results collectively demonstrate that the proposed semantic alignment and multimodal fusion strategies contribute not only to improved detection performance on benchmark datasets but also to enhanced generalizability in real-world scenarios. The model adapts effectively to different domains, capturing violent behaviors while suppressing erroneous detections in non-violent settings. Such robustness is essential for practical applications in surveillance and safety-critical environments, where the ability to generalize across diverse conditions determines the true value of an anomaly detection system.

5 Conclusion

This study addresses the problem of weakly supervised anomaly detection in surveillance videos by proposing a potent multimodal fusion framework that effectively integrates spatial, temporal, acoustic, and structural human motion cues. Our objective was to improve the robustness and precision of abnormal event detection through better feature alignment and adaptive fusion mechanisms.

This was achieved by introducing the MFMS for semantic-level alignment and the GMF mechanism for context-aware integration. These strategies, combined with the addition of pose modality and Contrastive Loss, allowed the model to mitigate modality discrepancies, prioritize informative signals, and enhance feature discriminability.

Our model achieved an AP of 84.98%, surpassing strong baselines. The results of ablation studies further validate the effectiveness of each component. specifically, the incorporation of pose features led to an improvement in AP from 83.51% to 84.34%. Replacing naive concatenation with GMF brought a performance boost of 0.64%. Furthermore, adopting Contrasive Loss instead of Triplet Loss yield an additional 0.64% gain. Finally, applying semantic alignment through MFMS contributed a further improvement of 0.78%. These findings collectively confirm that each proposed component plays a crucial role in enhancing detection accuracy and promoting better cross-modal feature alignment.

Nevertheless, certain limitations remain. Training convergence tends to plateau after extended epochs, and the model shows sensitivity to anomalies lacking clear human activity cues. These observations suggest that while our fusion framework enhances detection capabilities, further improvements are possible.

Future work will focus on deeper ablation to isolate the contributions of MFMS and GMF independently, incorporating dynamic loss reweighting strategies, and exploring adaptive temporal scaling for improved generalization. Additionally, real-time optimization will be investigated to facilitate deployment in practical surveillance systems.

References

[1] Rabia Ali, Muhammad Umar Karim Khan, and Chong Min Kyung. "Self-supervised representation learning for visual anomaly detection". In: *arXiv preprint arXiv:2006.09654* (2020).

[2] Antonio Barbalau et al. "SSMTL++: Revisiting self-supervised multi-task learning for video anomaly detection". In: *Computer Vision and Image Understanding* 229 (2023), p. 103656.

[3] Rui Z Barbosa and Hugo S Oliveira. "A Unified Approach to Video Anomaly Detection: Advancements in Feature Extraction, Weak Supervision, and Strategies for Class Imbalance". In: *IEEE Access* (2025).

[4] Zhe Cao et al. "OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43.1 (2021), pp. 172–186. DOI: 10.1109/TPAMI.2019.2929257.

[5] Joao Carreira and Andrew Zisserman. "Quo vadis, action recognition? a new model and the kinetics dataset". In: *proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2017, pp. 6299–6308.

[6] Rizwan Chaudhry et al. "Histograms of oriented optical flow and binet-cauchy kernels on nonlinear dynamical systems for the recognition of human actions". In: *2009 IEEE conference on computer vision and pattern recognition*. IEEE. 2009, pp. 1932–1939.

[7] Jian Cheng et al. "A multi-stage fusion instance learning method for anomalous event detection in videos". In: *International Journal of Machine Learning and Cybernetics* 14.2 (2023), pp. 445–454.

[8] Konstantinos Demertzis et al. "A Cross-Modal Dynamic Attention Neural Architecture to Detect Anomalies in Data Streams from Smart Communication Environments". In: *Applied Sciences* 13.17 (2023), p. 9648.

[9] Alexey Dosovitskiy et al. "An image is worth 16x16 words: Transformers for image recognition at scale". In: *arXiv preprint arXiv:2010.11929* (2020).

[10] Haodong Duan et al. *Revisiting Skeleton-based Action Recognition*. 2021. arXiv: 2104.13586 [cs.CV].

[11] Haodong Duan et al. "Revisiting Skeleton-based Action Recognition". In: *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022, pp. 2959–2968. DOI: 10.1109/CVPR52688.2022.00298.

[12] Ayush Ghadiya et al. "Cross-Modal Fusion and Attention Mechanism for Weakly Supervised Video Anomaly Detection". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 1965–1974.

[13] Ayush Ghadiya et al. "Cross-Modal Fusion and Attention Mechanism for Weakly Supervised Video Anomaly Detection". In: *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 2024, pp. 1965–1974. DOI: 10.1109/CVPRW63382.2024.00202.

[14] Mahmudul Hasan et al. "Learning temporal regularity in video sequences". In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016, pp. 733–742.

[15] Shawn Hershey et al. "CNN architectures for large-scale audio classification". In: *2017 ieee international conference on acoustics, speech and signal processing (icassp)*. IEEE. 2017, pp. 131–135.

[16] Wenping Jin, Li Zhu, and Jing Sun. "Aligning First, Then Fusing: A Novel Weakly Supervised Multimodal Violence Detection Method". In: *arXiv preprint arXiv:2501.07496* (2025).

[17] Byoung Chul Ko, Kwang-Ho Cheong, and Jae-Yeal Nam. "Fire detection based on vision sensor and support vector machines". In: *Fire Safety Journal* 44.3 (2009), pp. 322–329.

[18] Pratibha Kumari and Mukesh Saini. "An adaptive framework for anomaly detection in time-series audio-visual data". In: *IEEE Access* 10 (2022), pp. 36188–36199.

[19] Eman I Abd El-Latif et al. "VGGish transfer learning model for the efficient detection of payload weight of drones using Mel-spectrogram analysis". In: *Neural Computing and Applications* 36.21 (2024), pp. 12883–12899.

[20] Chun-I Lee et al. "End-to-end deep learning for recognition of ploidy status using time-lapse videos". In: *Journal of Assisted Reproduction and Genetics* 38.7 (2021), pp. 1655–1663.

[21] Zhian Liu et al. "A hybrid video anomaly detection framework via memory-augmented flow reconstruction and flow-guided frame prediction". In: *Proceedings of the IEEE/CVF international conference on computer vision*. 2021, pp. 13588–13597.

[22] Peng Lu et al. *RTMO: Towards High-Performance One-Stage Real-Time Multi-Person Pose Estimation*. 2024. arXiv: 2312.07526 [cs.CV]. URL: https://arxiv.org/abs/2312.07526.

[23] Weixin Luo, Wen Liu, and Shenghua Gao. "A revisit of sparse coding based anomaly detection in stacked rnn framework". In: *Proceedings of the IEEE international conference on computer vision*. 2017, pp. 341–349.

[24] Chengqi Lyu et al. *RTMDet: An Empirical Study of Designing Real-Time Object Detectors*. 2022. arXiv: 2212.07784 [cs.CV]. URL: https://arxiv.org/abs/2212.07784.

[25] Phong Thanh Nguyen et al. "Deep Learning based Optimal Multimodal Fusion Framework for Intrusion Detection Systems for Healthcare Data." In: *Computers, Materials & Continua* 66.3 (2021).

[26] Xiaogang Peng et al. *Learning Weakly Supervised Audio-Visual Violence Detection in Hyperbolic Space*. 2023. arXiv: 2305.18797 [cs.CV].

[27] Claudio Piciarelli, Christian Micheloni, and Gian Luca Foresti. "Trajectory-based anomalous event detection". In: *IEEE Transactions on Circuits and Systems for video Technology* 18.11 (2008), pp. 1544–1554.

[28] Bernhard Schölkopf et al. "Support vector method for novelty detection". In: *Advances in neural information processing systems* 12 (1999).

[29] Sandeep Singh Sengar, Abhishek Kumar, and Owen Singh. "Efficient Human Pose Estimation: Leveraging Advanced Techniques with MediaPipe". In: *arXiv preprint arXiv:2406.08021* (2024). URL: https://arxiv.org/abs/2406.08021.

[30] Waqas Sultani, Chen Chen, and Mubarak Shah. "Real-world anomaly detection in surveillance videos". In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018, pp. 6479–6488.

[31] Shengyang Sun and Xiaojin Gong. "Multi-scale bottleneck transformer for weakly supervised multimodal violence detection". In: *2024 IEEE International Conference on Multimedia and Expo (ICME)*. IEEE. 2024, pp. 1–6.

[32] Wenwen Sun et al. "Multimodal and multiscale feature fusion for weakly supervised video anomaly detection". In: *Scientific Reports* 14.1 (2024), p. 22835.

[33] Weijun Tan, Qi Yao, and Jingfeng Liu. "Overlooked video classification in weakly supervised video anomaly detection". In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2024, pp. 202–210.

[34] Yu Tian et al. "Weakly-supervised video anomaly detection with robust temporal feature magnitude learning". In: *Proceedings of the IEEE/CVF international conference on computer vision*. 2021, pp. 4975–4986.

[35] Zhan Tong et al. "Videomae: Masked autoencoders are data-efficient learners for self-supervised video pre-training". In: *Advances in neural information processing systems* 35 (2022), pp. 10078–10093.

[36] Guodong Wang et al. "Video anomaly detection by solving decoupled spatio-temporal jigsaw puzzles". In: *European Conference on Computer Vision*. Springer. 2022, pp. 494–511.

[37] Dong-Lai Wei et al. "Look, listen and pay more attention: Fusing multi-modal information for video violence detection". In: *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2022, pp. 1980–1984.

[38] Donglai Wei et al. "MSAF: Multimodal supervise-attention enhanced fusion for video anomaly detection". In: *IEEE Signal Processing Letters* 29 (2022), pp. 2178–2182.

[39] Peng Wu et al. "Not only look, but also listen: Learning multimodal violence detection under weak supervision". In: *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXX 16*. Springer. 2020, pp. 322–339.

[40] Fisher Yu and Vladlen Koltun. "Multi-scale dilated convolutional networks for vision". In: *Proceedings of the IEEE International Conference on Computer Vision*. 2015, pp. 393–401.

[41] Chen Zhang et al. "Exploiting completeness and uncertainty of pseudo labels for weakly supervised video anomaly detection". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023, pp. 16271–16280.

[42] Xuhong Zhang et al. "Gradient norm clipping: An empirical study". In: *arXiv preprint arXiv:1905.11881* (2019).

[43] Zhengming Zhao et al. "Hybrid selective harmonic elimination PWM for common-mode voltage reduction in three-level neutral-point-clamped inverters for variable speed induction drives". In: *IEEE Transactions on Power Electronics* 27.3 (2011), pp. 1152–1158.